

Multi-Layer Acoustic & Linguistic Feature Fusion for ComParE-23 Emotion and Requests Challenge

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ABSTRACT

The ACM Multimedia 2023 ComParE challenge focuses on classification/regression tasks for spoken customer-agent and emotionally rated conversations. The challenge baseline systems build upon the recent advancement in large-scale supervised/unsupervised foundational acoustic models that demonstrate consistently good performance across tasks. In this work, with the aim of improving the performance further, we present a novel multi-layer feature fusion method. In particular, the proposed approach leverages the hierarchical information from acoustic models using multi-layer statistics pooling, where we compute the weighted sum of layer-wise (mean and standard deviation) features. We further experimented with linguistic features and their late fusion with acoustic features, especially for subtasks involving complex conversations. Exploring various combinations of methods and features, we present four different systems tailored for each subchallenge, demonstrating significant performance gains over the baseline on the development and test set.

CCS CONCEPTS

- **Information systems** → **Information systems applications**;
- **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Computational Paralinguistics; Emotion Recognition; Request Categorization; Acoustic Modeling

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1 INTRODUCTION

This work presents our systems for the ACM Multimedia 2023 Computational Paralinguistics (ComParE) Challenge [9]. ComParE

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challenge consists of two subchallenges: 1) Emotion Share and 2) Request subchallenge, where the performance in the case of the former one is measured in terms of Spearman's correlation while for the latter performance is measured in terms of Unweighted Average Recall (UAR). The baseline systems provided by the organisers employ either popular openSMILE acoustic feature set [4] or embeddings extracted from neural models such as autoencoders or Wav2Vec class of models[1, 11]. More details can be found in Section 4 and baseline paper [9].

The challenge baseline results demonstrate that for the Emotion subchallenge, Wav2Vec2 model-based system performed significantly better than other systems. Building further, we improved the baseline system with 17% relative improvement on the development set using multi-layer acoustic feature fusion. In particular, we employ multi-layer statistics pooling where we compute the weighted sum of layer-wise (mean and standard deviation) features.

In the case of the Requests subchallenge, the baseline Wav2Vec2 based system performed the best on the Request task and the second best (but comparable) on the Complaint task. The best system for the Complaint task uses the openSMILE acoustic feature set denoted as ComParE hereafter in this paper. On the Request task, we improved over the baseline with a 23% relative improvement on the development set. This improvement was achieved using a language-specific (French) pretrained Wav2Vec2 model instead of the baseline English model and the multi-layer feature fusion approach described above. On the Complaint task, this strategy gave only marginal gains, which can be attributed to complex linguistic interaction during the conversation. To address this, we used a pretrained language model on the Automatic Speech Recognition (ASR) generated audio transcriptions. With recent advancements in ASR models, a lot of speech classification problems can be brought into the domain of the Natural Language Processing (NLP) domain by converting the speech into corresponding text transcription. We leverage this to construct another way to process and classify speech. Since the acoustic feature contains complementary information, we performed a late fusion of features from the respective acoustic and language models resulting in our best system on the Complaint task with a 22% relative improvement on the development set. The experiments presented in this work provide important insights into the importance of acoustic/linguistic cues, the choice of pretrained models and efficient feature fusion suitable for individual tasks. Our future work will extend the exploration to data-dependent/independent analysis of model's learning behavior via tools such as those presented in [5, 6].

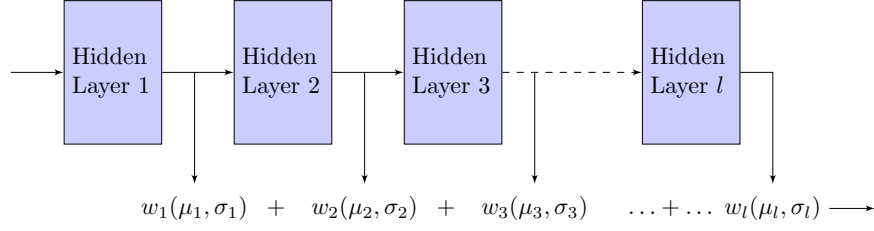


Figure 1: Multi-layer statistics pooling for acoustic feature fusion

2 STATISTICS POOLING BASED MULTI-LAYER ACOUSTIC FEATURE FUSION

Conventionally, for a downstream task, segment-level embeddings from a pretrained deep acoustic neural model's encoder (feature extractor) are extracted using average pooling applied over the output of the hidden layer. Recent works have shown that incorporating second-order statistics (standard deviation) along with mean embedding can greatly boost the performance of the system [7]. Further, while embeddings are extracted from the last or penultimate encoder layer, earlier layers can also encode useful acoustics cues in a hierarchical manner [2, 5]. We combined these two strategies for effective temporal modelling of speaker traits such as emotion.

In this work, we focus on the transformer models based on which our systems are developed. Assigning a learnable weight w_l to transformer hidden layer h_l the multi-layer acoustic feature F is computed as:

$$F = \frac{\sum_{l=1}^L e^{w_l} f_l}{\sum_{l=1}^L e^{w_l}} \quad \text{s.t. } f_l = \mu_l \oplus \sigma_l \quad (1)$$

$$\mu_l = \frac{\sum_{i=1}^T h_l^i}{T}, \quad \sigma_l = \sqrt{\frac{\sum_{i=1}^T (h_l^i)^2}{T} - \mu_l^2},$$

where $\oplus$ denotes the concatenation operator, L is the number of layers and μ_l & σ_l represent the temporal mean and standard deviation over T timesteps of each layer output. The weighted sum of the layerwise features uses the softmax function to ensure that weights are non-negative and they sum to 1. This can be interpreted as a sort of attention over layers; however, such attention is usually computed using a separate neural network comprising a few linear layers, something worth exploring in future work. Another line of investigation can be the use of attentive statistics pooling, where attention is applied over temporal frame-level features at a hidden layer output.

3 PROPOSED SYSTEMS

3.1 System-I (Emotion Subchallenge)

System-I is based on the same baseline Wav2Vec2 model that is fine-tuned on the MSP-Podcast [10] emotion dataset. We incorporated the proposed multi-layer feature fusion method to fine-tune the model for emotion recognition on the train split of the challenge dataset. The fine-tuned model (discarding the classification head) is used to extract the features, which are then fed to an SVM regressor.

3.2 System-II (Requests Subchallenge)

3.2.1 System-II.a (Acoustic Model). This system is based on the pretrained XLSR Wav2Vec2 model on French¹. The extracted embeddings from the pretrained model are used to train a binary SVM classifier.

3.2.2 System-II.a1 (Acoustic Model). This system is based on the pretrained XLSR Wav2Vec2 model on French¹. Similar to the Emotion subchallenge, we incorporated the multi-layer feature fusion method to fine-tune the model on the train split of the challenge dataset. In this case, we didn't use any external classifier.

3.2.3 System-II.b (Linguistic Model). This system uses the efficient & fast pretrained DistilBERT² [8] a distilled version of the BERT base multilingual model for linguistic features and XLSR Wav2Vec2 model on French¹ for generating ASR transcriptions and multi-layer acoustic features. Audio transcriptions from both customer and agent are used separately to fine-tune the DistilBERT model. Following the original BERT architecture [3], we extract the '[CLS]' embeddings for customer and agent transcriptions and concatenate them together before passing them to a classifier to evaluate the performance on the development set.

3.2.4 Linear Projection. For both System-II.a1 & II.b, we empirically found that using a single linear layer to convert the feature vector into logits instead of multiple layers had a small but positive boost to the model performance, e.g., boosting the recall by 1% on Request task.

3.2.5 System-II.c (Late Fusion). We also experimented with multi-modality in the form of late fusion to investigate the complementary nature of individual feature pipelines of System-II.a1 & II.b. We froze both the acoustic and linguistic models and trained a 2-layer MLP classifier on the concatenated embeddings.

4 DATASET DESCRIPTION

The ACM Multimedia 2023 Computational Paralinguistics Challenge has two subchallenges. The Hume-Prosody dataset for the Emotion Share subchallenge is provided by Hume AI. The dataset for the Request subchallenge has been provided by STIH Laboratory, Sorbonne University. A brief description of the tasks and datasets is provided below for better readability of this work.

¹<https://huggingface.co/jonatasgrosman/wav2vec2-large-xlsr-53-french>

²<https://huggingface.co/distilbert-base-multilingual-cased>

Table 1: Performance on ‘Development’ & ‘Test’ set in terms of Spearman’s correlation (ρ) of various systems for the Emotion subchallenge

System	Dev (ρ)	Test (ρ)
Challenge Baseline (Wav2Vec2)	0.500	0.5140
System-I (Ours)	0.587	0.5709

Table 2: Performance on ‘Development’ set in terms of UAR (%) of various systems for the Requests subchallenge

System	UAR	
	Request Task	Complaint Task
Challenge Baseline (Wav2Vec2)	65.1%	50.9%
Challenge Baseline (ComParE)	60.9%	52.9%
System-II.a (Ours)	67%	53%
System-II.a1 (Ours)	80%	50%
System-II.b (Ours)	77%	60%
System-II.c (Ours)	75%	61.4%

Table 3: Performance on the ‘Test’ set in terms of UAR (%) of various systems for the Requests subchallenge

System	UAR	
	Request Task	Complaint Task
Challenge Baseline (Wav2Vec2)	67.1%	52.2%
Challenge Baseline (ComParE)	58.8%	52.9%
System-II.a1 (Ours)	81.9%	-
System-II.c (Ours)	-	68.1%

4.1 Emotions Share

The Emotions Share subchallenge deals with a multi-label logistic regression task on speech utterances. The challenge corpus consists of short multilingual recordings, with each recording categorized by multiple speakers to belong to none, some, or all of 9 emotion classes: ‘Anger’, ‘Boredom’, ‘Calmness’, ‘Concentration’, ‘Determination’, ‘Excitement’, ‘Interest’, ‘Sadness’, and ‘Tiredness’. Further, the number of speakers who rate it as a particular emotion is denoted to be the label of the recording for that emotion. The final label for each recording is thus a 9D-normalized vector with each entry between 0 and 1.

4.2 Requests

The Requests subchallenge deals with the multi-label classification of customer-agent calls in a Customer Relationship Management (CRM) scenario. Each call has two labels for it: 1) (Complaint task) whether the customer has a complaint or not; 2) (Request task) the type of customer request: either concerning membership or an organization process, for example, a reimbursement. Each 30sec call has two channels where; the first channel corresponds to the customer’s audio, and the second corresponds to the agent’s audio. Both the raw dual-channel audio as well as the normalized mono-channel audios are available for analysis.

5 TRAINING SETUP

All of our models were trained using mixed precision on 3× Nvidia RTX-3090 GPUs for 50 epochs with a batch size of 3, ADAM optimizer, learning rate of 10^{-5} & linear warmup performed for the first 100 steps. We used cross-entropy loss for the Requests subchallenge and an elementwise sum of binary cross-entropy for the Emotion subchallenge. For training SVMs (wherever applicable), we used the same procedure as reported in the challenge baseline.

All the experiments are reproducible & implementation is available on GitHub³

6 EXPERIMENTAL RESULTS AND DISCUSSION

6.1 Emotion Subchallenge

The results of this experiment are reported in Table2. It can be observed that the proposed multi-layer acoustic fusion leads to a significant boost in performance for this task with 17% and 11% relative improvement over the baseline on the development and test set. We attribute this to our method’s ability to exploit and combine the hierarchical information in hidden layers efficiently. We believe other attention mechanisms, such as temporal, channel, or both, can further improve the performance of the system.

³<https://github.com/Cross-Caps/ComParE23>

6.2 Requests Subchallenge

The results of this experiment are reported in Table 1. Firstly, observe that by just replacing the backend pretrained model (French from English), System-II.a performed better than the challenge baseline system. This highlights the importance of a language-specific model in order to efficiently capture both acoustic and linguistic cues from audio, e.g., a model trained in English is not able to capture low-level phonemes and speech patterns unique to French. For more generic (multilingual/codemix) scenarios, we suggest the choice of pretrained model should be governed by the amount of mismatch between training and test data.

For the Request task, observe that our multi-layer acoustic fusion-based System-II.a.1 achieves the best performance with a 23% relative improvement over the baseline on the development set. On the Complaint task, we observed purely acoustic models achieve performance comparable to baseline, suggesting the need to model additional complex linguistic information for this task. To support our claims, observe that DistilBERT-based linguistic System-II.b alone achieves 17% relative improvement over the baseline. System-II.c achieves the best result for the Complaint task, which fuses both the acoustic and linguistic information with a 22% relative improvement over the baseline on the development set. Interestingly, while System-II.b & System-II.c did improve over the baseline; they were not better than acoustic-only System-II.a, something we would like to investigate further in future. Consistent with results on the development set, System-II.a.1 & System-II.c achieved significant improvement on the test set providing compelling evidence in support of the proposed approach.

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